

AI Code Review Assistant

The Problem

Manual code review is slow and inconsistent, and issues (bugs, security gaps, style drift) are often caught only after code is pushed or during a PR review — later than ideal. Developers want fast, structured feedback on their own changes before they commit, without waiting on a teammate or a CI pipeline.

The Solution

A local-first AI code review tool that hooks into git. It reads the staged diff before a commit, sends it to an LLM through a provider-agnostic adapter layer with a review prompt, and returns inline, line-mapped comments — bugs, security risks, readability issues, and style suggestions — directly to the developer, before the code ever leaves the machine.

Key Features

- **Local git integration** — intercepts the staged diff via a git pre-commit hook, runs before the commit completes
- **GitHub API mode** — optional alternate input source that pulls a diff from a GitHub PR (v2 / stretch goal)
- **Inline, line-mapped comments** — LLM output parsed to file + line number + comment, shown next to the relevant code
- **Selectable review depth** — checkboxes for Bugs, Security, Readability, and Style — user picks which categories to run
- **Multi-language support** — user selects which language(s) are in the diff so the prompt can be tuned accordingly
- **Provider adapter layer (OpenRouter now, pluggable later)** — LLM calls go through a common adapter interface (send diff + prompt, get structured comments back); OpenRouter is the only implementation built this week, but the interface is designed so other providers (OpenAI, Anthropic, local models) can be added later without touching the review logic
- **Review report view** — simple local UI (CLI output and/or lightweight web view) to read results per commit

Constraints — What It Should NOT Do

- **Does not auto-fix or auto-commit code** — it flags issues; the developer decides what to fix
- **Does not block commits (initially)** — no blocking of commits by default in week 1 — warn only, don't hard-block the workflow
- **Does not analyze the entire codebase** — all review happens on the local diff; full-repo or historical analysis is out of scope this week
- **No GitHub integration in v1** — GitHub PR fetching, OAuth, and webhook triggers are deferred to a later phase
- **No multi-user / cloud features** — no accounts, cloud sync, or team dashboards this week — single-user, local tool only
- **Only one provider implemented in v1** — the adapter interface is provider-agnostic by design, but only the OpenRouter adapter is built and tested this week — other providers are future work
- **No built-in API key storage/UI** — the tool assumes the user supplies their own OpenRouter key; no key management UI beyond a config file